

2026 PNU TECH WEEK · PHYSICAL AI HACKATHON

Autonomous Search & Rescue

A TurtleBot3 that maps an apartment, finds two red apples, delivers them to the safe zone and drives home on its own.

Team Alien

챗제이야 · 찻수뽀이 · 텃까웅산 · 손엘래나



WHO WE ARE

Team Alien



첢제이아



첢수똥이



텃까웅산



손엘래나

Built with

Webots R2025a

TurtleBot3 Burger

Python

OpenCV

NumPy

Ultralytics YOLO11n

THE CHALLENGE

Rescue two red apples

- 1 Find 2 red apples on the floor
- 2 Confirm each one up close
- 3 Deliver to the safe zone
- 4 Drive back to the start

WHAT MAKES IT HARD

- Sensors only: no GPS, no ground truth
- Decoys: green, purple and orange apples, a red can
- A pedestrian walks through the rooms



The Webots apartment world the robot has to search

OUR CONTRIBUTION

What we were given, what we built

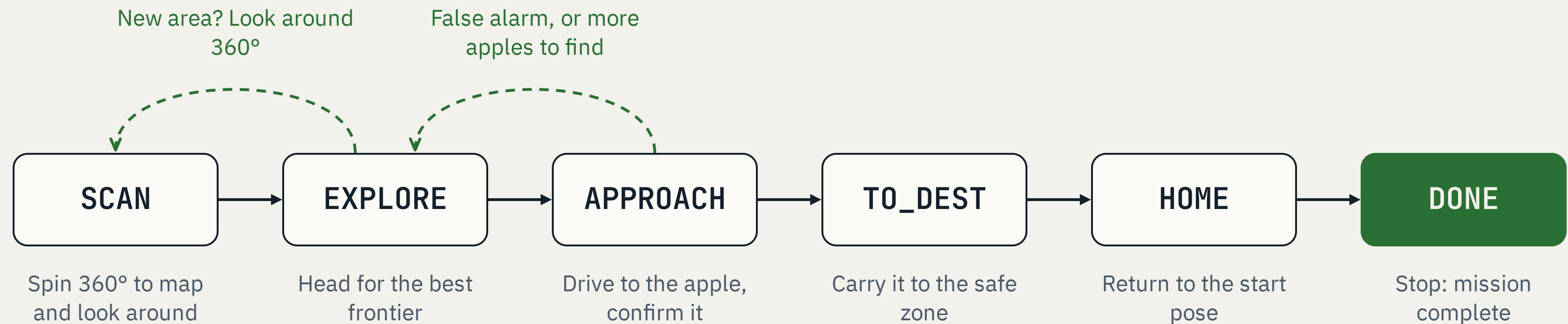
GIVEN • STARTER KIT

Keyboard driving demo
Sensor and camera demos
Color segmentation demo
YOLO11n model + demo
Ground-truth pose viewer
Apartment world + apples
Breakroom test worlds

BUILT BY US • `tb3_sar` • ~1,200 lines

- ✓ Mission state machine (6 states)
- ✓ Compass calibration + scan matching
- ✓ Occupancy grid + camera coverage map
- ✓ Semantic object map + frontier scoring
- ✓ Red-apple detector with 6 checks
- ✓ A* + pure pursuit + DWA planner
- ✓ Safety layer + stuck recovery
- ✓ Live map, camera and costmap views

Six states, one mission



Safety layer, every step: something within 20 cm → stop, back off, re-plan.

HOW IT WORKS

Inside the controller

SENSE

Wheel encoders

Compass

360° LiDAR

Camera

THINK

Localize

odometry + compass +
LiDAR scan matching

Perceive

red-apple detector + YOLO
object map

Map

log-odds occupancy grid, 5
cm cells

Plan

frontier choice + A* path

ACT

Pure pursuit

DWA planner

Wheel speeds

Safety stop

Knowing where we are

01

Odometry + compass

Wheels measure distance; the compass, calibrated on the first spin, gives heading.

02

LiDAR scan matching

Nudge the pose up to ± 15 cm until the scan lines up with known walls.

03

Occupancy grid

An 18×18 m map with 5 cm cells, updated from every LiDAR scan.

4.49 m

of position correction applied by 247 scan-match fixes during our final run.

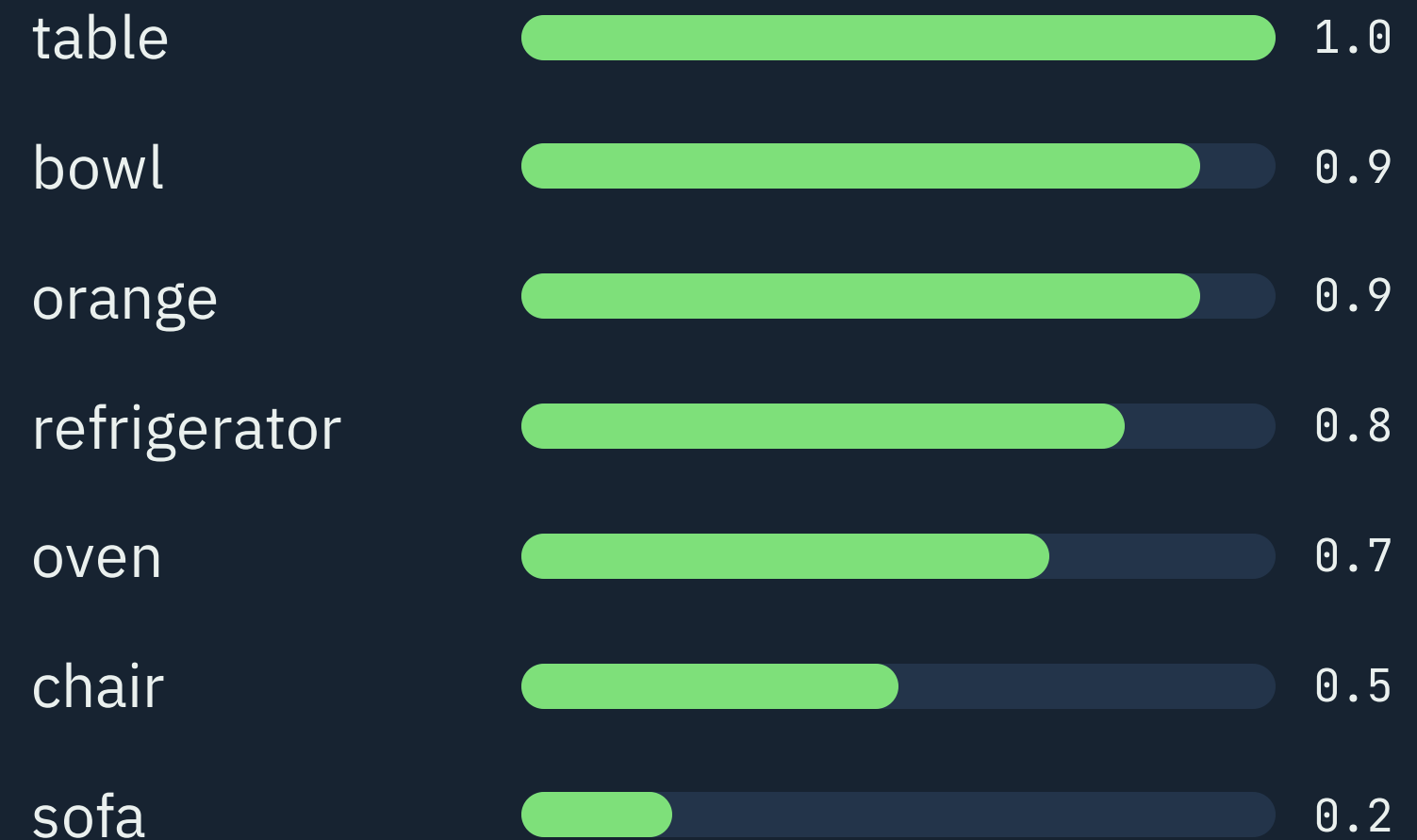
Where would an apple be?

- 01 Frontiers are map edges and floor the camera hasn't checked yet.
- 02 YOLO names objects; the LiDAR range pins them on the map.
- 03 Pick the frontier that is close and near kitchen-like things.

$\text{score} = \text{path} - 3.0 \times \sum \text{prior} \times \text{share} \times e^{(-d / 1.5 \text{ m})}$

Lowest score wins. In our run it picked $(-0.3, -9.6)$ over the nearest frontier $(-1.1, -10.6)$.

Apple prior per object



Is it really a red apple?

01 **Red** — inside the red hue band

02 **Round** — circular, close to square

03 **On the floor** — below the horizon line

04 **Right size** — fits its distance on the floor

05 **Whole** — not cut off at the image edge

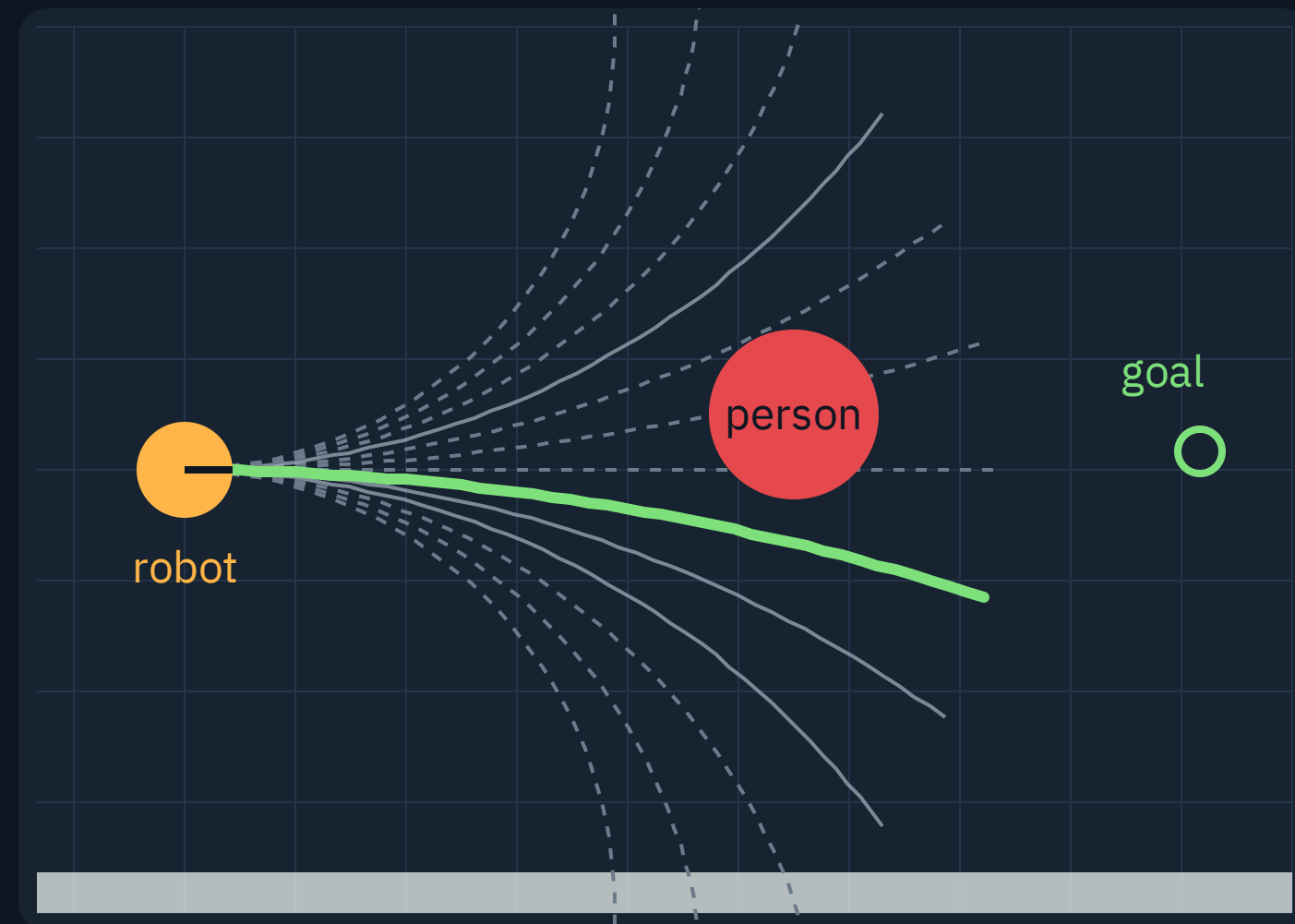
06 **Not tall** — no red continuing above it

3 matching sightings before the robot switches to APPROACH

2 close-up confirmations under 2 m before an apple counts

0 false alarms in our final run, despite four decoys

Driving safely around people



Grey: safe options · dashed: would collide · green: chosen

GLOBAL · A*

A* on a map where walls are padded by the robot's size; re-planned every 1.5 s.

LOCAL · DWA

Tests 90 trajectories 1.5 s ahead on a live 3 m costmap, so people leave no ghosts.

SAFETY · every step

Under 20 cm: stop, wait, back off, re-plan. No progress in 3.5 s: recover.

FINAL RUN

Mission complete in 502.5 s

2 / 2

red apples found and confirmed

0

false alarms

2

times stuck, recovered on its own



Search for apple #1

0 → 298.2 s

Search for apple #2

298.2 → 430.4 s

Deliver to safe zone

430.4 → 479.5 s

Drive home

479.5 → 502.5 s

WHAT'S NEXT

Built in, ready to switch on



YOLO-World detector

An open-vocabulary model for apartment objects. Our final run used the YOLO11n fallback.



Jev frontier chooser

TypeSafe AI's Jev picks among the top 6 frontiers in a background thread; the local prior is the fallback.



Football mode

Set TARGET_NAME to “football” to hunt a soccer ball with YOLO instead of the color detector.

TEAM ALIEN

Thank you

Questions?

챗제이야 · 찿수똥이 · 텃까웅산 · 손엘래나

github.com/Xuf313/hackathon